

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.27078 -0.74879 -0.06417  0.63512  1.74067

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.006795   0.425485   7.067 7.41e-08 ***
hp           0.021684   0.002635   8.229 3.48e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.006 on 30 degrees of freedom
Multiple R-squared:  0.693,    Adjusted R-squared:  0.6827
F-statistic: 67.71 on 1 and 30 DF,  p-value: 3.478e-09
```

Some of the individual components from the fitted model are listed below for reference:

```
> m$residuals
    Mazda RX4    Mazda RX4 Wag   Datsun 710   Hornet 4 Drive
0.608014994  0.608014994   -1.023364771    0.608014994
Hornet Sportabout  Valiant    Duster 360    Merc 240D
1.198584681  0.716432710   -0.319263348   -0.351174929
...

> m$model
              cyl  hp
Mazda RX4         6 110
Mazda RX4 Wag     6 110
Datsun 710         4  93
Hornet 4 Drive    6 110
Hornet Sportabout  8 175

> m$terms
cyl ~ hp
attr(,"variables")
list(cyl, hp)
attr(,"factors")
  hp
cyl 0
hp  1
attr(,"term.labels")
[1] "hp"
attr(,"order")
[1] 1
attr(,"intercept")
[1] 1
attr(,"response")
[1] 1
attr(,".Environment")
<environment: R_GlobalEnv>
```

```
attr(,"predvars")
list(cyl, hp)
attr(,"dataClasses")
      cyl      hp
"numeric" "numeric"

> m$sqr
$qr
              (Intercept)      hp
Mazda RX4             -5.6568542 -829.78980772
Mazda RX4 Wag           0.1767767  381.74189579
Datsun 710              0.1767767    0.12620114
Hornet 4 Drive          0.1767767    0.08166844
Hornet Sportabout       0.1767767  -0.08860368
...

> confint(m)
              2.5 %      97.5 %
(Intercept) 2.13783849 3.87575200
hp           0.01630186 0.02706522

> anova(m)
Analysis of Variance Table

Response: cyl
      Df Sum Sq Mean Sq F value    Pr(>F)
hp      1  68.517   68.517   67.71 3.478e-09 ***
Residuals 30  30.358    1.012
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We can again use the *predict()* function to determine the number of cylinders required for a given horsepower. In the following example, we observe that in order to achieve 200 horsepower, we require at least 7 cylinders.

```
> hp <- data.frame(hp = 200)
> cyl <- predict(m, hp)
> cyl
      1
7.343504
```

You are encouraged to read the manual pages for the above R functions to learn more on their arguments, options and usage. 

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The author is a free software enthusiast.